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Quantitative modelling of shipping freight rates: developments in the past 20 years

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ABSTRACT

Understanding the determination mechanism and dynamics of shipping freight rates has significant implications for practitioners, economists, and regulators of maritime transport. This paper systematically reviews the relevant papers on shipping freight rate published between 2000 and 2019. Specifically, it provides empirical insights including the major influencing factors of shipping freight rates and differences among the shipping sectors, and discusses the dominant methods and their applicability, strengths and weaknesses, and opportunities for new methods. Furthermore, the omissions and limitations in methods are explored to identify the research gaps and set a future research agenda. It is found that there is a strong focus on temporal dynamics with a wide range of modelling methods applied, while discussion on cross-sectional dynamics on how shipping freight rates are determined by vessel characteristics, contract terms, market competition, etc., is limited. Besides, the collaboration network analysis shows that more knowledge sharing and cross-discipline collaboration efforts can create potential for more inspired insights with advanced methods in this field.

KEYWORDS

Ocean shipping; freight rate; quantitative modelling; time series; cross-sectional dynamics; collaboration network

1. Introduction

Ocean transportation is the cornerstone of globalization and the main contributor of international transportation networks that enable international trade and underpin global supply chains. As one of the most important elements of the ocean shipping market, shipping freight rates play an important role in regulating the supply and demand of sea transport, and directly affect revenues and profits of ocean carriers.

The term ‘freight rate’ is widely used in ocean shipping studies. It can be generally defined as the ‘*amount of money paid to a shipowner or shipping line for the carriage of each unit of cargo (tonnage, cubic meter, or container load) between named ports*’ (Stopford 2009). However, it often means different subjects across the ocean shipping sectors and contract types. The data and methods applied in the literature also reflect the different fundamental understandings of the definition of ‘freight rate’ in ocean shipping. Notably, for the bulk shipping sectors (i.e., dry bulk and tanker), the freight rate is essentially the vessel hiring price (formally called as vessel fixture rate), which can be a time-limited hiring (a time charter [TC] contract) or a mission-limited hiring (a voyage charter contract). In contrast, the container shipping sector has a clear two-level market structure, that is,

the ‘upper’ containership charter market where the time-charter rate is paid by the charterers to the shipowners, and the ‘lower’ freight market where the shipment price is paid by the shippers to the carriers. Furthermore, in the shipping freight derivatives market, rates of Forward Freight Agreements (FFAs) (traded either as freight futures or freight options) are used (Stopford 2009), although strictly, these are not freight rates.

Accurate explanation and forecasting of the shipping freight rates have significant implications for practitioners, economists, and regulators. The annual standard deviation of the Baltic Capesize Index (BCI) was almost 40% from March 1999 to October 2011, more than twice as high as the volatility of the Morgan Stanley Capital International (MSCI) World Stock Market Index (Drobetz, Richter, and Wambach 2012). Stopford (2009) identified 22 dry cargo freight cycles between 1741 and 2007. The volatile and cyclical shipping freight rates bring great challenges to shipowners, charterers, and shippers to successfully plan their business, and to shipping investors to hedge and speculate based on freight rate changes.

The overarching goal of the research on shipping freight rates should be to improve the understanding of the dynamic behaviours of freight rates, as well as their determination. However, due to the lack of comprehensive reviews, it remains unclear how much improvement has been made towards this goal, as well as challenges and gaps. There have been literature reviews of shipping market research. Alexandridis et al. (2018) summarized papers on freight rate volatility and volatility spillovers across shipping segments when conducting an extensive review on shipping finance research. Two reviews specifically on shipping freight rates were published in 2006, by Glen (2006) and Kavussanos and Visvikis (2006), focusing on model development in the bulk shipping market and the shipping derivatives market, respectively. As the number of papers published has increased greatly, especially after 2010, an updated review is highly needed.

For this purpose, this paper systematically reviews the academic articles on quantitative models of shipping freight rates published between 2000 and 2019 across all three shipping sectors with four general research objectives. Following the guidance of Wee, Bert and Banister (2016) for review studies, this paper aims to provide added values to the literature by (i) providing empirical insights on shipping freight rates, including what factors have been proven to have an impact on shipping freight rates, what are the major differences of freight rates among different shipping sectors; (ii) analysing the dominant methods and their applicability, strengths and weaknesses, and opportunities for new methods; and (iii) exploring the omissions and limitations in existing literature to define the gaps in the literature and set a future research agenda.

The remainder of this paper is organized as follows: the second section introduces the review methodology and an overview of the studies. In the third section, the reviewed papers are categorized into four research objective groups, followed by a discussion of the quantitative models under each group. The fourth and fifth sections thoroughly discuss main findings and challenges regarding freight rate influencing factors and time series characteristics, respectively. Additionally, the sixth section provides an analysis of collaboration patterns in this area. The last section presents the conclusions of this paper.

2. Research method

The relevant studies are searched and selected through an iterative process of defining appropriate searching terms, reviewing the literature, conducting initial analysis, and finalizing classification results. The details of the search methodology are as follows:

2.1. Search methodology

A computerized search is used to collect journal publications on shipping freight rates. The database Web of Science is first searched with the keywords ‘freight rate’, ‘freight market’, ‘spot rate’, ‘charter rate’, ‘Baltic Dry Index (BDI)’, ‘Baltic Dirty Tanker Index (BDTI)’, ‘Baltic Clean Tanker Index

(BCTI)', 'Shanghai Containerized Freight Index (SCFI)', and 'China (Export) Containerized Freight Index (CCFI)' combined with 'shipping', 'maritime', 'dry bulk', 'container', and 'tanker'.¹ The same search is then expanded to different publishers' websites, including Elsevier (www.sciencedirect.com), Springer (link.springer.com), Taylor & Francis (www.tandfonline.com), Emerald (www.emerald.com/insight/), and JSTORE (www.jstore.org). Totally 468 articles were located for the time between 2000 and 2019.

2.2. Inclusion/exclusion criteria

This review considers peer-reviewed academic articles studying shipping freight rates that were written in English over the last 20 years, from 2000 to 2019. Other forms of publications, such as conference papers, books, reports, and articles published outside this period are excluded. Considering the fact that those important and foundational studies on this topic published before 2000 have been well reviewed in Glen (2006) and Kavussanos and Visvikis (2006), this paper, therefore, focuses on the developments in freight rate research after 2000.

To set a clear research boundary, the main emphasis is on quantitative studies that analyse characteristics of shipping freight rates, examine the impact of macroeconomic and shipping-related factors on the shipping freight rates, forecast shipping freight rates, or investigate economic spillovers between the shipping freight market and other markets. Specifically, studies related to shipping freight rates are excluded if their focus is on other topics. For example, some studies used freight rate as an independent variable to analyse other markets. After filtering out the articles that are not peer-reviewed academic papers, the abstracts of the remaining articles were manually inspected to further filter out the irrelevant papers based on the inclusion/exclusion criteria, 168 articles were finally selected, and were thoroughly reviewed and analysed.

2.3. Basic overview and next steps towards in-depth analysis

After collecting the final 168 papers, the review process begins with an overview to obtain some general and basic observations. The next step is to categorize the studies into a few groups based on their research objectives and summarize and compare the analysis methods for each research purpose. The major research findings and progresses are also discussed by research groups and shipping sectors. Finally, we identify the research trends and gaps. This helps set future research directions synchronously across the research objectives and shipping sectors.

Figure 1 demonstrates the number of published papers each year, along with the development of the dry bulk shipping market represented by the BDI. The number of papers shows a fast-increasing trend, reflecting the increasing research interest starting around 2011, following the 2008–2009 financial crisis and slowdown in ocean transportation. Deteriorated shipping market conditions seem to have increased attention to the importance of freight rate risk management and a better understanding of the shipping freight rates.

The selected articles were published in 50 different international journals in the fields of transportation and economics, particularly in maritime economics. Table 1 shows a detailed list of paper distribution in journals. Seventeen journals published over 80% of the selected articles, with three journals contributing over 45%: Maritime Economics & Logistics, followed by Maritime Policy & Management, and Transportation Research Part E: Logistics and Transportation Review.

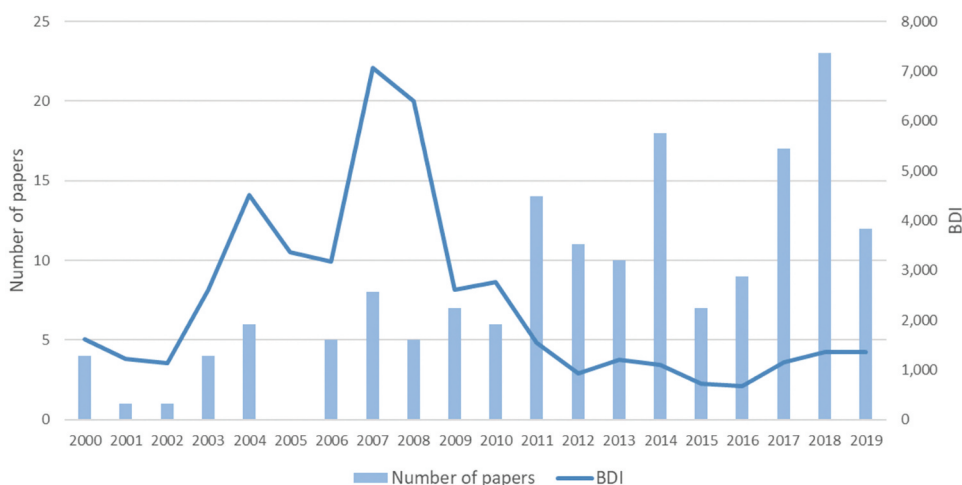


Figure 1. Distribution of articles per year versus BDI (Source of BDI: Clarksons Research 2020).

Table 1. Distribution of articles in journals.

Journal	Number of papers	Percentage
Maritime Economics & Logistics	26	15.48%
Maritime Policy & Management	25	14.88%
Transportation Research Part E: Logistics and Transportation Review	25	14.88%
The Asian Journal of Shipping and Logistics	8	4.76%
Journal of Transport Economics and Policy	6	3.57%
Physica A-Statistical Mechanics and Its Applications	6	3.57%
International Journal of Shipping and Transport Logistics	6	3.57%
Energy Economics	5	2.98%
Applied Economics	5	2.98%
International Journal of Transport Economics	5	2.98%
Transportation Research Part A: Policy and Practice	5	2.98%
Expert Systems with Applications	3	1.79%
Explorations in Economic History	2	1.19%
International Journal of Forecasting	2	1.19%
Economic Modelling	2	1.19%
Journal of Transport Geography	2	1.19%
Journal of Futures Markets	2	1.19%
Top 17 journals total	135	80.36%
Other 33 journals	33	19.64%

3. Research objectives and models

This section starts with a categorization of the reviewed papers based on their research objectives, followed by a discussion of the general choices of methods across research objectives and shipping sectors.

3.1. Research objectives

The research objectives of the reviewed papers are divided into the following four groups:

- (1) *analysing time series characteristics of shipping freight rates,*
- (2) *analysing the factors affecting the temporal dynamics of shipping freight rates,*
- (3) *forecasting shipping freight rates, and*
- (4) *analysing the factors affecting the cross-sectional dynamics of shipping freight rates.*

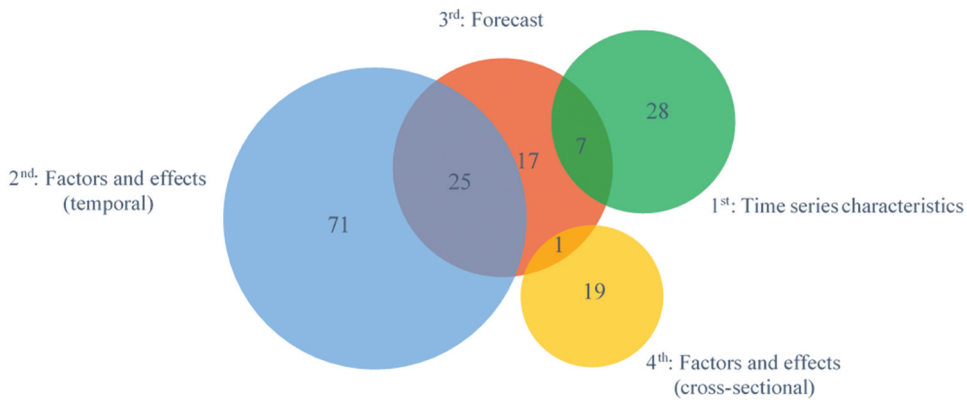


Figure 2. Number of papers per research objectives.

The categorization is based on two kinds of dynamics in freight rates: one is temporal dynamics, containing time-based patterns and trends (related to objective groups (1), (2), and (3)); and the second is cross-sectional dynamics that describes changes in freight rates caused by external factors irrespective of time (related to objective groups (4)). Studies on temporal dynamics involve papers that (1) analyse time series characteristics including stationarity, seasonality, cyclic patterns, etc., usually using univariate models and focusing on the freight rate itself; (2) analyse external factors affecting freight rates over time; and (3) forecast future freight rates. Studies on cross-sectional dynamics of freight rates are categorized as objective group (4). This group has similarity to group (2) as both investigate the influence of external factors on shipping freight rates. They are analysed separately due to the difference in research methods (i.e., time series analysis and non-time series analysis). Furthermore, while papers in group (2) often involve macroeconomic factors and study spillover effects between different markets, papers in group (4) capture the influencing factors at a micro-level such as trading routes, contract types, and vessel characteristics (Adland, Cariou, and Charles Wolff 2017).

Figure 2 presents numbers of papers in each objective group. Specifically, the topics of the first group (35 papers) include seasonality, stationarity, or independent and identical distribution of shipping freight rates. This group also includes papers that attempted to detect bubbles, identify shipping cycles, and study data volatility clustering characteristics.

The second group (96 papers) attempts to estimate the factors affecting the temporal dynamics of shipping freight rates or analyse the spillover effects between the shipping freight markets and other markets, such as the newbuilding market and commodity market, or examine the relationship between different shipping freight markets.

The third group has 33 papers that aim to forecast freight rates using univariate or multivariate models.

The fourth group involves 20 papers that investigate the factors affecting the cross-sectional dynamics of freight rates. Although micro-level fixture data are widely available, only six papers used individual fixture data to conduct micro-level analysis and evaluate the vessel and contract-specific factors. This group also includes 12 papers that study trade and port-specific effects on container freight rates and two papers studying macro-level factors (e.g., ship price and commodity price) of dry bulk freight rates without considering the time effect.

Clearly, the focus of the existing shipping freight modelling literature is on the temporal dynamics of the shipping freight rates, in comparison to the cross-sectional dynamics. This preference of research focus is also directly reflected in data choices. Approximately 93% papers used highly aggregated data as freight rate measures, that is, either freight rate indices (about 50%) or quasi-index data (about 43%) that were processed based on relevant market fixtures and presented the market average for each vessel type and/or route at each time. Furthermore, only

few studies used individual fixture data (e.g., Adland, Cariou, and Charles Wolff 2016) or container freight rate data directly collected from shipping lines in container studies to analyse trade and port-specific factors (e.g., Wilmsmeier and Hoffmann 2008; Lei and Bachmann 2019).

3.2. Quantitative models

3.2.1. Overview of models

A wide variety of quantitative models have been applied to shipping freight rate analysis, as summarized in Table 2. The mostly used models were econometric methods (75%), including the relatively new and popular autoregressive conditional heteroskedasticity (ARCH)-type models, vector autoregressive (VAR), and vector error correction models (VECMs). Besides, equilibrium model that explain the behaviour of supply, demand, and freight rates and other data-driven mathematical approaches such as empirical mode decomposition, multifractal detrended fluctuation analysis, etc. that are designed to handle nonstationary and nonlinear data, were also used for freight rate studies. Moreover, machine learning methods applied to shipping freight rate analysis were only limited to artificial neural network (ANN) and support vector machine (SVM) algorithms.

3.2.2. Models by research objectives

Breaking down the applications based on research objectives, approximately one-third papers (54) dealt with the shipping freight rates per se using univariate models, and the other two-thirds (116 papers) analysed factor impacts using multivariate models. There was a sharp contrast between the bulk sector versus the container sector in terms of univariate model applications (about 64% vs. 9%) which is mostly due to the data quality varieties in each sector.

The dry bulk sector has the earliest and most varieties of daily indices, which are regularly published by international shipping organizations, mainly the Baltic Exchange. In contrast, the container sector had only weekly indices, CCFI and SCFI, both based on containerized export trade from China. Only after 2020, the daily global container freight index—the Freightos Baltic Index (FBX) became available (Baltic Exchange 2020). Furthermore, there has been no charter rate index for the container sector.

The univariate econometric models are either used for time series characteristics analysis or forecasting. The traditional econometric models applied here are autoregressive moving average (ARMA), autoregressive integrated moving average (ARIMA), and ARCH-type models. Other nonlinear and nonparametric methods such as empirical mode decomposition (EMD), multifractal detrended fluctuation analysis (MF-DFA), rescaled range analysis, stochastic differential equation

Table 2. Summary of common methods.

Method		First introduced in	Number of papers
Classic econometric model	Linear Regressions	1886	14
	Autoregressive moving average (ARMA)	1951	6
	Autoregressive integrated moving average (ARIMA)	1970	15
	ARCH-type	1982	37
	VAR	1980	33
	VECM	1995	25
	Structural equation model (SEM)	1960s	3
Theoretical model	Equilibrium model	1870s	8
Simulation	System dynamics (SD) simulation	1958	5
Other	Empirical mode decomposition (EMD)	1998	7
	data-driven		
	Multifractal detrended fluctuation analysis (MF-DFA)	2002	5
	mathematical approach		
	Rescaled range analysis	1951	7
Machine learning	Stochastic differential equation (SDE)	1942	6
	Wavelet transform	1984	3
	Artificial neural network (ANN)	1964	12
	Support vector machine (SVM)	1995	1

(SDE), and wavelet transform are mainly used for the analysis of time series characteristics (e.g., cyclic behaviour, volatility cluster, and stationarity) of freight rates, as they have the advantage of being unlimited by linearity and stationarity.

Forecasting papers predominantly focused on the freight rate temporal dynamics in the dry bulk and tanker sectors, most often using the dataset published by the Baltic Exchange. Only three papers attempted to forecast the container freight rates (Luo, Fan, and Liu 2009; Nielsen et al. 2014; Munim and Joachim Schramm 2017). Model performance comparisons were conducted in 20 papers, which were summarized in Table 3. Besides the traditional econometric methods (e.g., ARIMA, GARCH, VAR, and VECM), machine learning was mostly used and were found in many studies to outperform the former types (Leonov and Nikolov 2012; Santos, Luciano, and Floriano 2014; Zeng and Chenrui 2014; Qingcheng et al. 2016). Despite the stronger predictive power of machine learning methods, they are normally not suitable for interpreting factor inter-relationships, which is often of great interest to shipping economists.

For factor analysis of the dynamic or static features, some of the multivariate econometric models applied include linear time series models focusing on dynamic factor interactions, linear non-time-series models focusing on static relationships, equilibrium models focusing on economic interpretation of the market driven factors. These models can be generally compared as data-driven models versus hypothesis-driven models. The representative models of the former include VAR models and VECMs which both can be modified to include contemporaneous effects and the latter is often used in cases of cointegrated variables. However, they do offer theoretical support and explanation for the choices of variables and lag lengths.

Alternatively, hypothesis-driven models, represented by the SEM and the equilibrium models, are easier to interpret, but require prior knowledge about the forces influencing variables and prior differentiation between endogenous and exogenous variables. Thus, they were much less used in shipping freight rate modelling compared to VAR models and VECMs. Equilibrium theory provides a good theoretical foundation for regression modelling, for example, 3-stage least squares (3SLS) estimation (Luo, Fan, and Liu 2009). Applying the equilibrium models to simulation allows an extensive analysis of freight rate dynamics and explicit scenario testing.

Finally, system dynamics (SD) simulation is suitable for complex dynamics systems that involve internal feedbacks and time delays. It also allows nonlinear relationships and can integrate stochastic processes to demonstrate the freight market behaviour. However, data requirements for SD simulations are intensive, restricting its extensive application in modelling shipping freight rates.

In the next two sections, the studies on influencing factors and time series characteristics of the shipping freight rates will be discussed in detail.

4. Determination of shipping freight rates

4.1. Disparity across shipping sectors

The three main shipping sectors, that is, dry bulk, tanker, and container, have very different characteristics and competitive market structures. While the bulk shipping market has been characterized as perfectly competitive, where similar ships compete for homogeneous cargoes (Stopford 2009), the container shipping sector is characterized with dynamic market competition conditions caused by frequent mergers and acquisitions, and an increasing market concentration level. Within the bulk shipping market, dry and wet (i.e., tanker) bulk markets also differ due to the very different commodities transported. Stopford (2009) compared dry cargo and tanker freight rates for the period 1947–2008 and showed that tanker freight rates had higher volatility. A potential explanation for this observation could be that commodity prices and freight rates are correlated and prices of liquid bulk such as crude oil are more volatile than prices of dry bulk such as wheat and steel. Therefore, it is important to investigate the differences among the shipping sectors and differentiate them while studying the freight rates.

Table 3. Forecast performance comparisons among models.

Paper	Forecast error measurement	RW	ARIMA	VAR	VECM	GARCH	EGARCH	TGARCH	APGARCH	KARCH	CGARCH	GJR-GARCH	GARCH-M	ARIMARCH	BPNN	DPN-BPNN	RBNN	DPN-RBNN	ELM	ANN	EDP-ANN	ANN-AGA	MNN	GFNN	PCANN	SOFM	MLP	SVR	FTS	FNN
Kavvasanos and Ntomikos (2003)	RMSE																													
Alizadeh et al. (2007)	RMSE, DWU																													
Batchelor et al. (2007)	RMSE																													
	daily BFT																													
Daru et al. (2010), and Uyar et al. (2016)	RMSE, MAE, NRMSE																													
	daily freight rate																													
Chen et al. (2012)	RMSE, MAE, Theil's																													
Daru (2012)	RMSE, NRMSE, MAE																													
Daru et al. (2012)	RMSE, NRMSE, MAPE																													
Goudos and Skindopoulos (2012)	MAE, RMSE																													
Ham et al. (2014)	RMSE																													
Leonov and Nikolov (2012)	RMSE																													
Lyridis et al. (2014)	squared error																													
Santos et al. (2014)	MAPE, RMSE																													
Zeng and Qu (2014)	MAPE, MAPE, RMSE																													
Zeng et al. (2016)	MAPE, MAPE, RMSE																													
Eslami et al. (2017)	RMSE																													
Munim and Schramm (2017)	MAE, MAE, MAPE																													
Tsioumas et al. (2017)	RMSE, RMSE																													
Giavittidis et al. (2018)	MAE, MAE																													
Zhang, Xue, and Stanley (2019)	MAPE, RMSE																													
	daily BDI																													
Zhang, Chen, Wang, Gu, and Stanley (2019)	MAPE, RMSE																													
	weekly BDI																													
	monthly BDI																													
	yearly BDI																													

Note: This table reports the forecast performance of each model used in different papers. Comparison should be done within each row, using the colour from green to yellow (or small to large numbers), where green (larger number) represents the better performing models and yellow (smaller number) represents the worse performing models. Forecast error measurements include mean absolute deviation (MAD), mean absolute error (MAE), mean absolute percentage error (MAPE), median relative absolute error (MdRAE), mean error (ME), mean percentage error (MPE), mean square Error (MSE), normalized root mean squared error (NRMSE), root mean squared error (RMSE), and standard deviation of error (SDE). Besides the model abbreviations defined previously, RW refers to random walk; EGARCH, Exponential GARCH; TGARCH, Threshold-GARCH, APGARCH, Asymmetric Power-ARCH; IGARCH, Integrated-GARCH; CGARCH, Component GARCH; GJR-GARCH, Glosten-Jagannathan-Runkle GARCH; and GARCH-M, GARCH-in-mean. ARIMARCH is a model developed in Munim and Joachim Schramm (2017) that combines ARIMA and ARCH. BPNN, RBNN, ELM, and DFN stand for back propagation neural network, radial basis function neural network, the extreme learning machine, and dynamic fluctuation network, respectively. AGA, adaptive genetic algorithm; MNN, modular neural network; GFNN, generalized feedforward neural network; PCANN, principal components analysis neural network; SOFM, self-organizing feature map; MLP, multi-layer perceptron; FTS, fuzzy time series; FNN, fuzzy neural network.

Most papers studied freight rates of dry bulk (62.5%) and tanker (29.2%), with only 13.1% on container. However, when only looking at the papers with multivariate models, the percentage share of papers on dry bulk decreases to 58.6%, while the percentage share of papers on tanker increases slightly to 29.3% and container to 17.2%. The commonly used influencing factors in the reviewed papers with numbers and percentage shares of papers with these factors are generalized by shipping sector in Table 4, categorized by shipping demand, shipping supply, and macroeconomic indicators. Seaborne trade volume for specific trade routes (in tons, TEUs, or ton-miles) and fleet capacity (in dead-weight tons or TEUs) are the most straightforward indicators for shipping demand and shipping demand, respectively.

A close inspection of Table 4 reveals that dry bulk and tanker sectors have many more common factor choices compared to the container sector. The special choices for the container shipping sector are extrapolated below:

- (1) Container sector analyses use the **seaborne trade volume** and **transport distance** as the demand indicators more often, compared to the other two sectors which mostly use commodity prices for freight rate studies. This is probably because the liner business serves too diversified cargo types to use a commodity price indicator.
- (2) For the container sector, factors representing service efficiency are often considered for freight rate analysis, as shippers in the container sector are the transport service receivers, instead of charterers who may manage the transport of cargos by themselves in the dry bulk and tanker sectors. Examples include **service frequency** (Márquez-Ramos et al. 2011; Suárez-Alemán, Campos, and Luis Jiménez 2015), **connectivity** (measured by the number of services [Márquez-Ramos et al. 2011; Wilmsmeier and Hoffmann 2008] or connectivity index [De Oliveira 2014]), and **port conditions**, such as port infrastructure (De Oliveira 2014; Wilmsmeier and Hoffmann 2008) and port utilization (Lei and Bachmann 2019).
- (3) Another unique factor for container freight rate analysis is **trade imbalance** (Wilmsmeier and Hoffmann 2008; Márquez-Ramos et al. 2011; De Oliveira 2014). This is reasonable because empty container reposition cost caused by trade imbalances can be significant for the liners. UNCTAD (2015) also reported that higher trade deficits lead to lower export container freight rates for a country.
- (4) Furthermore, **market competition** has also been included specifically for the container sector, as the container sector has experienced frequent mergers and acquisitions, establishment, and reconstruction of shipping alliances, and increasing market concentration level. Several studies have found the market competition in the container shipping sector, measured by the number of carriers (Suárez-Alemán, Campos, and Luis Jiménez 2015; Wilmsmeier and Hoffmann 2008) and market concentration index (Hirata 2017; De Oliveira 2014), has a significantly negative impact on container freight rates.
- (5) Finally, two papers included **maximum vessel size** (Gouvernal and Slack 2012; Wilmsmeier and Hoffmann 2008) in container freight rate model. Increasingly larger container ships have been built over the past decades to reduce unit operating cost. To verify this effect empirically, both papers examined and confirmed an inverse relationship between the maximum vessel size in the market and freight rate. Different from the container sector, the studies on bulk freight rates investigated vessel charter prices that allow reference to vessel sizes explicitly, instead of being limited to the largest ship size in market.

The disparity of the influencing factor choices across the shipping sectors comes from their different operation characteristics including different cargo forms, transporting, and handling equipment, and organization approaches. Most dry bulk and tanker carriers carry raw or semi-raw materials with solid or liquid forms organized based on each individual shipment, while

container shipping carriers generally use containers and complicated liner services to transport manufactured goods.

Another notable finding from Table 4 is that the tanker sector often uses the number of ships in the market, under demolition, and new delivery to represent the supply, while dry bulk studies use a more general supply factor, that is, fleet capacity. This is because of the more designated vessels in the tanker sector, which makes it possible to focus on a special cargo type and tanker type. Additionally, macroeconomics indicators (stock and job market indicators) are usually used only for the dry bulk sector, as major dry bulk commodities are the major raw materials for production inputs and are commonly accepted as an indicator of the overall macroeconomic developments.

Although the studies appear to agree on the disparity across shipping sectors, the appropriate influencing factors for each freight rate measurement remains an evolving topic. For example, Bai and Siu Lee Lam (2017) showed that using shipping demand in ton-miles influences the number of ships, which further affects freight rates in the very large gas carrier (VLGC) shipping market. However, Jane Jing, Leung Yip, and Liu (2011) revealed that freight rates Granger cause new-building prices, and rejected the directional relationship from newbuilding prices to freight rates using both VECMs and VAR models. Frederik and Nomikos (2019) used AIS data to generate supply estimates of the fleet supply condition. Using VECM, the authors found that the freight rate evolution depends on the fleet's geographical distribution and its employment status.

In addition to the general economic factors listed in Table 4, the interactions of different types of freight rates were studied. For example, 27 papers analysed the interactions of spot rates with FFA rates, three papers on spot rates and TC rates, and five papers on across market sectors. A general conclusion has been made that FFA rates contribute to the discovery of new information in the spot market.

Furthermore, some freight rate influencing factors that are considered in the container liner shipping practices are not yet discussed extensively in the scientific literature. For example, port congestion and empty container shortage have been the consequences of COVID-19 pandemic,

Table 4. Summary of common factors.

Category	Variable	Frequency of factors					
		Dry bulk (44 papers)		Tanker (26 papers)		Container (17 papers)	
Demand	Seaborne trade volume	5	11.4%	6	23.1%	8	47.1%
	General trade volume	4	9.1%	2	7.7%	1	5.9%
	Transport distance	1	2.3%	1	3.8%	6	35.3%
	Commodity production	4	9.1%	3	11.5%	0	0.0%
	Commodity prices	8	18.2%	17	65.4%	1	5.9%
Supply	Trade imbalance	0	0.0%	0	0.0%	3	17.6%
	Fleet capacity	14	31.8%	4	15.4%	5	29.4%
	Number of ships	1	2.3%	3	11.5%	1	5.9%
	Ship order	3	6.8%	3	11.5%	1	5.9%
	Ship delivery/newbuild	0	0.0%	5	19.2%	1	5.9%
	Ship demolition	1	2.3%	7	26.9%	1	5.9%
	Maximum vessel size	0	0.0%	0	0.0%	2	11.8%
	Ship price	6	13.6%	4	15.4%	1	5.9%
	Ship age	4	9.1%	2	7.7%	0	0.0%
	Market competition	0	0.0%	1	3.8%	4	23.5%
	Port condition	1	2.3%	0	0.0%	4	23.5%
	Connectivity	0	0.0%	0	0.0%	3	17.6%
	Service frequency	0	0.0%	0	0.0%	2	11.8%
	Oil price	11	25.0%	17	65.4%	5	29.4%
	Stock market indicator	9	20.5%	2	7.7%	1	5.9%
Macroeconomic indicators	GDP	3	6.8%	0	0.0%	3	17.6%
	Industrial production	4	9.1%	1	3.8%	0	0.0%
	Interest rate	2	4.5%	3	11.5%	0	0.0%
	Currency exchange rate	4	9.1%	0	0.0%	0	0.0%

Note: Only papers with multivariate models are counted in this table.

which are believed to be responsible for the rapid and persistent growths in CCFI and SCFI since mid-2020. Besides, US dollar exchange rate could affect container freight rates, as currency adjustment factor is charged to offset losses for carriers from fluctuating exchange rates between the US dollar and other currencies.

4.2. Influence of vessel characteristics and contract terms

In addition to the factors listed in Table 4, it is worth noting that a limited number (seven) of papers have investigated vessel specifications and contract details' influence on freight rates. Specifically, Adland, Cariou, and Charles Wolff (2017) found that around 70–80% of rate variation is explained by the time fixed effects while the remainder is due to vessel characteristics, contractual terms, and the influence of charterers in an analysis of the Offshore Support Vessel market.

Alizadeh and Talley (2011; 2011b) investigated the influence of the laycan period, the vessel's hull type (for tanker), deadweight, vessel age, and voyage routes. Roar and Alizadeh (2018) considered vessel specifications and contractual terms, and Adland, Cariou, and Charles Wolff (2016) and Adland, Cariou, and Charles Wolff (2017) focused on the influence of the characteristics of charterers and ship owners. Another two papers that worth notice are Köhn and Thanopoulou (2011) and Agnolucci, Smith, and Rehmatulla (2014) which specifically considered fuel efficiency effect, in addition to vessel and charter term characteristics. The latter found that fuel efficiency leads to a freight rate premium for the Panamax dry bulk TC market during 2008–2012. Furthermore, ship owners recoup approximately 40% of the fuel savings through higher rates.

5. Time series characteristics of shipping freight rates

5.1. Stationarity

Despite the long time and intensive research focus on time series analysis of shipping freight rates, it is still debatable whether freight rates are truly non-stationary. Tvedt (2003), Adland and Cullinane (2006), Koekebakker, Adland, and Sødal (2006), Kou, Luo, and Zhao (2018), and Población (2017) contributed in-depth analyses and discussions about the stationarity of bulk freight rates in levels. They all mentioned that the equilibrium freight rate resulting from the interaction between supply and demand should be mean reverting (stationary), as the bulk shipping freight market is a (close to) perfectly competitive market (Stopford 2009).

Specifically, Tvedt (2003) found evidence on the stationary nature of freight rates (weekly dry bulk spot rate and BFI) by changing the perspective from US\$ to Japanese yen. Koekebakker, Adland, and Sødal (2006)'s research results of weekly spot and forward freight rates of five tanker routes are also in line with this theory in a non-linear setting. Adland and Cullinane (2006)'s empirical findings suggested that the spot rate in the tanker sector is locally non-stationary over the range of the process. However, the existence of a non-linear mean-reverting drift at the edges of the range is sufficient to pull the series back into its middle region and determine the global stationarity. Kou, Luo, and Zhao (2018) drew the same conclusion from observations in the dry bulk sector that long-term freight rate samples (between 26 to 42 years) are stationary. However, merely increasing the sample length would not make the freight rates stationary, while sample windows separated by structural breaks are tested as stationary.

Yet, with only a few exceptions (e.g., SCFI monthly freight indexes on different routes in Jiang, Jian, and Gong 2018), most reviewed articles have treated shipping freight rates in levels as non-stationary based on common stationarity tests like Augmented Dickey–Fuller (ADF), Phillips–Perron (PP), and Kwiatkowski–Phillips–Schmidt–Shin (KPSS) tests. Therefore, most papers have taken the first differences, or the first logarithmic differences of shipping freight rates. However, this may have been a problematic practice considering that over-differencing may cause many issues in

parameter estimation, for example, biased estimates of the effects of long-memory processes (Hurvich and Chen 2000).

5.2. Seasonality and cycles

A few papers have systematically investigated the seasonal behaviour of freight rates. Kavussanos and Alizadeh-M (2001, 2002) and Yin and Shi (2018) found deterministic seasonality in the dry bulk, tanker, and container sectors, respectively, while Poblacion (2015) concluded that seasonality in bulk freight rates is stochastic. Población and Serna (2018) found that the seasonal period is approximately one year in the tanker market.

There are ten papers analysing the shipping cycles. Using the same freight rate data and applying rescaled range analysis with V-statistic, Goulielmos (2010) detected medium cycles of 10 and 20 years. This result is in line with Stopford (2009), who identified 22 dry cargo freight cycles between 1741 and 2007. Using the same method as Goulielmos (2010) and more recent data (i.e., 1-year time charter rate from 1989 to 2008), Goulielmos and Elpiniki Psifia (2011) found a short term 4-year cycle. Chen et al. (2019) found the short-term cycle of the market is 41.2 months (i.e., 3.43 years), the medium-to-long-term cycles are 7.11 years to 8.97 years, and the long-term cycle is 11.96 years for the Aframax tanker market using wavelet analysis. Chisté and van Vuuren (2014) extracted a longer 7-year cycle and a shorter 4-year cycle from the BDI using Fourier analysis. Moreover, Yijie, Yin, and Sheng (2018) found that the shipping cycles have impacts on the causality relationship among the freight rate, second-hand price, and new-build ship price in the dry bulk shipping market. The causality could only be identified during the trough and peak periods.

5.3. Volatility clustering

Volatility clustering is another important characteristic of the time series of shipping freight rates. ARCH-type models are often used as they can capture the heteroskedasticity of many freight rate data and examine how variables influence the mean and variance of the shipping freight rates. However, Engelen et al. (2011) argued that nonlinearities, the rejection of normality, and excess kurtosis and skewness indicate the misleading nature of ARCH-type models with respect to stationarity. Therefore, many researchers chose nonparametric specifications (e.g., rescaled range analysis and MF-DFA) that allow one to determine the correlation properties of both non-stationary and stationary time series. Specific examples are Chen et al. (2016) and Tingyi et al. (2018) that used MF-DFA to study the volatility of dry bulk and tanker freight rates, respectively; Liu and Chen (2018) applied MF-DFA to investigate the asymmetric volatility of the dry bulk shipping market under structural breaks. However, these nonparametric approaches in the literature were mostly limited to univariate analysis, while multivariate GARCH models were also used to investigate the volatility spillover effects among different markets.

6. Research collaborations

Shipping freight rate research requires deep knowledge about the shipping industry background, and certain analytical and modelling skills. In the previous sections, discussions around methods and research objectives indicated the existence of certain gaps and controversy in the literature. To seek for the reasons for underdeveloped topics, it is meaningful to see how knowledge was shared and what collaboration network structure exists for shipping freight rate analysis. The collaboration networks among authors and institutes in the past 20 years for shipping freight rate analysis (Figures 3 and 4) are created using the Fruchterman Reingold layout in Gephi 0.9.2 software. The size of the nodes represents the number of papers, and the thickness of edges represents the number of collaborations.

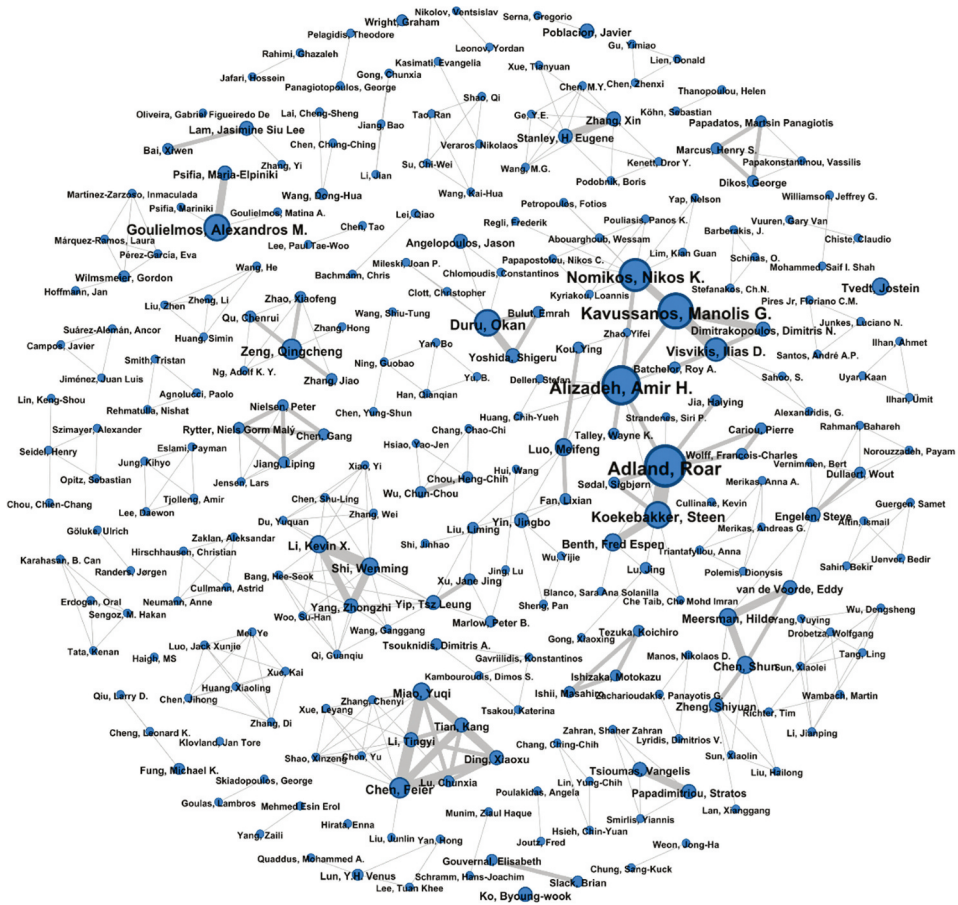


Figure 3. Co-authorship network (Nodes and edges are weighted according to the frequency of appearance).

Overall, the authorship network is widely dispersed with 1.64 papers/author on average. Approximately 71% authors have only one published paper in this field. However, there are several major networks that made significant contributions to this topic: the network led by Alizadeh, Nomikos, Kavussanos, and Visvikis has 24 papers in total; the network led by Adland and Koekebakker has 14 papers, followed by two networks with 7 published papers each led by Goulielmos and Duru; another two networks have 5 papers each and are led by Feier Chen and Zeng. These six networks contribute over 36% of all papers.

Four major scholars on this topic emerge: Adland (12 papers), Alizadeh (11 papers), Kavussanos (10 papers), and Nomikos (9 papers). The latter three had intensive collaboration among themselves. And their works were more linked to financial analysis, which is also partly shown by journal selections. Unsurprisingly, their papers mainly focused on the time series characteristics or the connections with the derivative markets of the bulk freight rates using time series models. But two studies by Alizadeh in collaboration with the shipping economist Talley in 2011 focused on the determinants of freight rates using contract details. Another clearly strong internal collaboration network is led by Feier Chen (5 papers), whose publications are mainly in statistics journals. Cross-group or even cross-field collaborations are rare.

Furthermore, as illustrated in Figure 4, the institutional collaboration networks show even more single nodes with much fewer edges, indicating the collaborations across institutions were not as often. Many co-authorship networks were within the same institutes, for example, Alizadeh,

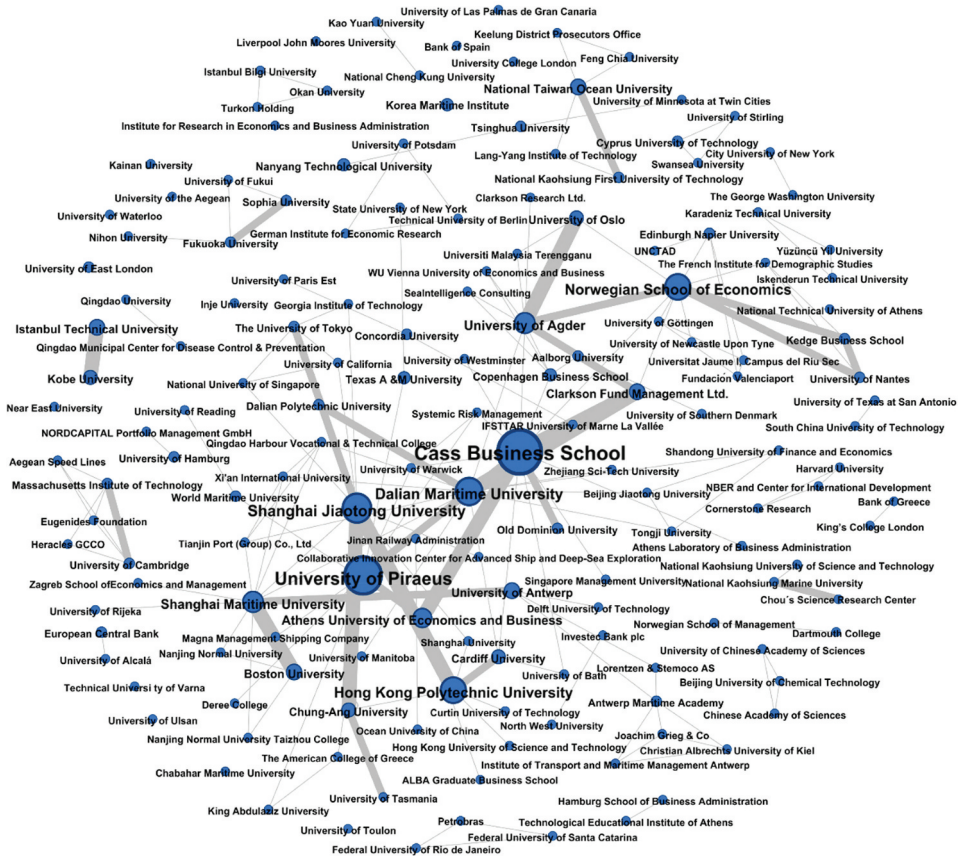


Figure 4. Institutional collaboration network (Nodes and edges are weighted according to the frequency of appearance).

Nomikos, and Kavussanos were all from the Cass Business School, and Feier Chen led a group of scholars in Shanghai Jiao Tong University. The figure also shows that shipping freight rates were studied by business schools (e.g., Cass Business School and the Norwegian School of Economics), or universities with a strong focus on maritime and transport studies (e.g., University of Piraeus, Dalian Maritime University, Shanghai Jiao Tong University, the Hong Kong Polytechnic University, and Shanghai Maritime University).

7. Conclusion

This review provides a comprehensive discussion of shipping freight rate research. In addition to exploring developments and limitations in the existing literature for the interest of shipping economists, it also intends to help shipping practitioners better understand the shipping freight market. For shipping speculators and investors interested in future freight rates developments, this paper compares different prediction models with different freight rate data and the main findings. For carriers and other shipping operators who are interested in maximizing profitability or risk control, this review provides a clear roadmap to the related research and thoroughly discusses the influencing factors that have been analysed in literature.

The main findings of this review are summarized below.

7.1. Trends and achievements

Only 22 papers (13.1% of the shipping freight rate papers) in the past 20 years studied container freight rates, 21 of which investigated the container liner shipping freight rates charged from shippers and the remaining one studied containership charter rates and the correlations among different shipping sectors.

In terms of research objectives, more focus was given to temporal dynamics compared to cross-sectional dynamics caused by micro-level factors (e.g., vessel characteristics, contract terms). However, even though intensively discussed, some topics regarding the temporal dynamic characteristics remain controversial, such as the stationarity issue.

In terms of model choices, a wide range of modelling methods are used. Overall, classic econometrics remains the most popular method, while machine learning algorithms only increased slightly since they first appeared in the shipping literature ten years ago. However, a general finding is that machine learning outperforms classic econometric methods in forecasting. Innovative hybrid models constructed by a few papers were also successful compared with the original models.

In terms of influencing factors, studies for the three shipping sectors have presented quite different findings, which accurately reflect the differences in the business operating models and cargo characteristics across sectors. In other words, the existing literature has reflected the well-established and consistent understanding of the shipping sector operating characteristics.

A further analysis of the authorship networks indicates that this is a very concentrated network with several major networks making significant contributions. Approximately 71% authors published only one paper on this topic. And six major institutions contributed to this topic with 27% of the authorship. There is a lack of cross-group, cross-field, or cross-institutional collaborations.

7.2. Gaps and future directions

First, compared to temporal analysis, cross-sectional analysis that could reveal the freight rate mechanism (e.g., explaining influences of operational factors on freight rates) is highly underdeveloped. These analyses could provide useful guidance to industrial practitioners for making better operational decisions. As discussed in [Section 4.2](#), the existing seven papers studying the influences of vessel characteristics and contract terms on freight rates have revealed important factor relationships, however, did not cover the entire shipping market. The cross-sectional analysis using fixture data can be extended to, for example, containership time charter rate, the Supramax market and the liquefied natural gas (LNG) market. In the container sector specifically, the impact of non-charter related cross-sectional factors such as port congestion, empty container availability and US dollar exchange rate on freight rates are considered in shipping practices but not yet investigated extensively in the scientific literature.

Second, as discussed earlier, when investigating the container freight rates, the focus was predominantly on shipping freight rates charged from shippers, while the containership charter rates as a major cost for the carriers remain largely overlooked. A better understanding of containership charter rates and the influencing factors is crucial for carriers to minimize cost and ship-owners to increase their revenue and capability in speculation.

Third, the (non)stationarity property of the shipping freight rates remains a controversial topic. The widely used log return methods are useful for shipping investors, but not for other shipping practitioners who are interested in the freight rate levels. Besides, over-differencing also causes issues in parameter estimation, for example, biased estimates of the effects of long-memory processes. With more high-quality data becoming foreseeably available, machine learning approach is anticipated to be increasingly applied due to its strong predictive power that is not limited by data stationarity. Additionally, despite its weaknesses in parameter

estimation and interpretability, theory about causal inference in machine learning is available (e.g., Pearl 2019).

Finally, current collaboration patterns indicate that knowledge sharing, and across-field collaboration efforts can be further improved, which will bring more inspired insights and advanced new methods to this topic. For example, a collaboration between shipping economists and computer scientists could help expedite the application of state-of-the-art machine learning algorithms in shipping freight rate studies and might enable the investigation of less-explored cross-sectional dynamics of shipping freight rates with increased interpretability.

Note

1. Papers studying the shipping freight derivatives market are also included.

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